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Gas sampling line

Abstract

A gas sampling line having a channel for conducting respiratory gases from a patient respiratory interlace to a gas monitor, the gas sampling line comprising, i.a., a gas sampling tube comprised of a polyether block amide material, the polyether segments of which comprise polyethyleneoxide. Use of a tube comprised of a polyether block amide material, the polyether segments of which comprise polyethyleneoxide, for sampling of respiratory gases; and a method for sampling of respiratory gases, the method comprising conducting respiratory gases through such a tube. A gas analysis system for analysing respiratory gases, comprising a gas sampling line as defined above and a gas monitor connectable to the gas sampling line.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4705543	12/1986	Kertzman	N/A	N/A
4960128	12/1989	Gordon et al.	N/A	N/A
4964408	12/1989	Hink et al.	N/A	N/A
5042500	12/1990	Norlien et al.	N/A	N/A
5319355	12/1993	Russek	N/A	N/A
5337744	12/1993	Branigan	N/A	N/A
5341805	12/1993	Stavridi et al.	N/A	N/A
D353195	12/1993	Savage et al.	N/A	N/A
D353196	12/1993	Savage et al.	N/A	N/A
5377676	12/1994	Vari et al.	N/A	N/A
D359546	12/1994	Savage et al.	N/A	N/A
5431170	12/1994	Mathews	N/A	N/A
5436499	12/1994	Namavar et al.	N/A	N/A
D361840	12/1994	Savage et al.	N/A	N/A
D362063	12/1994	Savage et al.	N/A	N/A
D363120	12/1994	Savage et al.	N/A	N/A
5456252	12/1994	Vari et al.	N/A	N/A
5479934	12/1995	Imran	N/A	N/A
5482036	12/1995	Diab et al.	N/A	N/A
5494043	12/1995	O'Sullivan et al.	N/A	N/A
5533511	12/1995	Kaspari et al.	N/A	N/A
5561275	12/1995	Savage et al.	N/A	N/A
5590649	12/1996	Caro et al.	N/A	N/A
5602924	12/1996	Durand et al.	N/A	N/A
5638816	12/1996	Kiani- Azarbayjany et al.	N/A	N/A

5638818	12/1996	Diab et al.	N/A	N/A
5645440	12/1996	Tobler et al.	N/A	N/A
5671914	12/1996	Kalkhoran et al.	N/A	N/A
5694922	12/1996	Palmer	N/A	N/A
5703161	12/1996	Steenblock et al.	N/A	N/A
5726440	12/1997	Kalkhoran et al.	N/A	N/A
D393830	12/1997	Tobler et al.	N/A	N/A
5743262	12/1997	Lepper, Jr. et al.	N/A	N/A
5747806	12/1997	Khalil et al.	N/A	N/A
5750994	12/1997	Schlager	N/A	N/A
5758644	12/1997	Diab et al.	N/A	N/A
5760910	12/1997	Lepper, Jr. et al.	N/A	N/A
5890929	12/1998	Mills et al.	N/A	N/A
5919134	12/1998	Diab	N/A	N/A
5987343	12/1998	Kinast	N/A	N/A
5997343	12/1998	Mills et al.	N/A	N/A
6002952	12/1998	Diab et al.	N/A	N/A
6010937	12/1999	Karam et al.	N/A	N/A
6027452	12/1999	Flaherty et al.	N/A	N/A
6040578	12/1999	Malin et al.	N/A	N/A
6066204	12/1999	Haven	N/A	N/A
6115673	12/1999	Malin et al.	N/A	N/A
6124597	12/1999	Shehada et al.	N/A	N/A
6128521	12/1999	Marro et al.	N/A	N/A
6129675	12/1999	Jay	N/A	N/A
6144868	12/1999	Parker	N/A	N/A
6152754	12/1999	Gerhardt et al.	N/A	N/A
6184521	12/2000	Coffin, IV et al.	N/A	N/A
6232609	12/2000	Snyder et al.	N/A	N/A
6241683	12/2000	Macklem et al.	N/A	N/A
6255708	12/2000	Sudharsanan et al.	N/A	N/A
6280381	12/2000	Malin et al.	N/A	N/A
6285896	12/2000	Tobler et al.	N/A	N/A
6308089	12/2000	von der Ruhr et al.	N/A	N/A
6317627	12/2000	Ennen et al.	N/A	N/A
6321100	12/2000	Parker	N/A	N/A
6334065	12/2000	Al-Ali et al.	N/A	N/A
6360114	12/2001	Diab et al.	N/A	N/A
6368283	12/2001	Xu et al.	N/A	N/A
6411373	12/2001	Garside et al.	N/A	N/A
6415167	12/2001	Blank et al.	N/A	N/A
6430437	12/2001	Marro	N/A	N/A
6430525	12/2001	Weber et al.	N/A	N/A
6463311	12/2001	Diab	N/A	N/A
6470199	12/2001	Kopotic et al.	N/A	N/A
6487429	12/2001	Hockersmith et al.	N/A	N/A
6505059	12/2002	Kollias et al.	N/A	N/A
6525386	12/2002	Mills et al.	N/A	N/A
6526300	12/2002	Kiani et al.	N/A	N/A

6534012	12/2002	Hazen et al.	N/A	N/A
6542764	12/2002	Al-Ali et al.	N/A	N/A
6580086	12/2002	Schulz et al.	N/A	N/A
6584336	12/2002	Ali et al.	N/A	N/A
6587196	12/2002	Stippick et al.	N/A	N/A
6587199	12/2002	Luu	N/A	N/A
6595316	12/2002	Cybulski et al.	N/A	N/A
6597932	12/2002	Tian et al.	N/A	N/A
6606511	12/2002	Ali et al.	N/A	N/A
6635559	12/2002	Greenwald et al.	N/A	N/A
6639668	12/2002	Trepagnier	N/A	N/A
6640116	12/2002	Diab	N/A	N/A
6640117	12/2002	Makarewicz et al.	N/A	N/A
6658276	12/2002	Kiani et al.	N/A	N/A
6661161	12/2002	Lanzo et al.	N/A	N/A
6697656	12/2003	Al-Ali	N/A	N/A
6697658	12/2003	Al-Ali	N/A	N/A
RE38476	12/2003	Diab et al.	N/A	N/A
RE38492	12/2003	Diab et al.	N/A	N/A
6738652	12/2003	Mattu et al.	N/A	N/A
6760607	12/2003	Al-Ali	N/A	N/A
6779522	12/2003	Smith et al.	N/A	N/A
6783573	12/2003	Richardson	N/A	N/A
6788965	12/2003	Ruchti et al.	N/A	N/A
6816241	12/2003	Grubisic	N/A	N/A
6822564	12/2003	Al-Ali	N/A	N/A
6850787	12/2004	Weber et al.	N/A	N/A
6850788	12/2004	Al-Ali	N/A	N/A
6876931	12/2004	Lorenz et al.	N/A	N/A
6920345	12/2004	Al-Ali et al.	N/A	N/A
6934570	12/2004	Kiani et al.	N/A	N/A
6943348	12/2004	Coffin IV	N/A	N/A
6956649	12/2004	Acosta et al.	N/A	N/A
6961598	12/2004	Diab	N/A	N/A
6970792	12/2004	Diab	N/A	N/A
6985764	12/2005	Mason et al.	N/A	N/A
6990364	12/2005	Ruchti et al.	N/A	N/A
6998247	12/2005	Monfre et al.	N/A	N/A
7003338	12/2005	Weber et al.	N/A	N/A
7015451	12/2005	Dalke et al.	N/A	N/A
7027849	12/2005	Al-Ali	N/A	N/A
D526719	12/2005	Richie, Jr. et al.	N/A	N/A
7096052	12/2005	Mason et al.	N/A	N/A
7096054	12/2005	Abdul-Hafiz et al.	N/A	N/A
D529616	12/2005	Deros et al.	N/A	N/A
7133710	12/2005	Acosta et al.	N/A	N/A
7142901	12/2005	Kiani et al.	N/A	N/A
7225006	12/2006	Al-Ali et al.	N/A	N/A
RE39672	12/2006	Shehada et al.	N/A	N/A
7228878	12/2006	Chen et al.	N/A	N/A

7254429	12/2006	Schurman et al.	N/A	N/A
7254431	12/2006	Al-Ali et al.	N/A	N/A
7254434	12/2006	Schulz et al.	N/A	N/A
7274955	12/2006	Kiani et al.	N/A	N/A
D554263	12/2006	Al-Ali et al.	N/A	N/A
7280858	12/2006	Al-Ali et al.	N/A	N/A
7289835	12/2006	Mansfield et al.	N/A	N/A
7292883	12/2006	De Felice et al.	N/A	N/A
7341559	12/2007	Schulz et al.	N/A	N/A
7343186	12/2007	Lamego et al.	N/A	N/A
D566282	12/2007	Al-Ali et al.	N/A	N/A
7356365	12/2007	Schurman	N/A	N/A
7371981	12/2007	Abdul-Hafiz	N/A	N/A
7373193	12/2007	Al-Ali et al.	N/A	N/A
7377794	12/2007	Al-Ali et al.	N/A	N/A
7395158	12/2007	Monfre et al.	N/A	N/A
7415297	12/2007	Al-Ali et al.	N/A	N/A
7438683	12/2007	Al-Ali et al.	N/A	N/A
7483729	12/2008	Al-Ali et al.	N/A	N/A
D587657	12/2008	Al-Ali et al.	N/A	N/A
7500950	12/2008	Al-Ali et al.	N/A	N/A
7509494	12/2008	Al-Ali	N/A	N/A
7510849	12/2008	Schurman et al.	N/A	N/A
7514725	12/2008	Wojtczuk et al.	N/A	N/A
7519406	12/2008	Blank et al.	N/A	N/A
D592507	12/2008	Wachman et al.	N/A	N/A
7530942	12/2008	Diab	N/A	N/A
7593230	12/2008	Abul-Haj et al.	N/A	N/A
7596398	12/2008	Al-Ali et al.	N/A	N/A
7606608	12/2008	Blank et al.	N/A	N/A
7620674	12/2008	Ruchti et al.	N/A	N/A
D606659	12/2008	Kiani et al.	N/A	N/A
7629039	12/2008	Eckerbom et al.	N/A	N/A
7640140	12/2008	Ruchti et al.	N/A	N/A
7647083	12/2009	Al-Ali et al.	N/A	N/A
D609193	12/2009	Al-Ali et al.	N/A	N/A
D614305	12/2009	Al-Ali et al.	N/A	N/A
7697966	12/2009	Monfre et al.	N/A	N/A
7698105	12/2009	Ruchti et al.	N/A	N/A
RE41317	12/2009	Parker	N/A	N/A
RE41333	12/2009	Blank et al.	N/A	N/A
7729733	12/2009	Al-Ali et al.	N/A	N/A
7761127	12/2009	Al-Ali et al.	N/A	N/A
7764982	12/2009	Dalke et al.	N/A	N/A
D621516	12/2009	Kiani et al.	N/A	N/A
7791155	12/2009	Diab	N/A	N/A
RE41912	12/2009	Parker	N/A	N/A
7880626	12/2010	Al-Ali et al.	N/A	N/A
7909772	12/2010	Popov et al.	N/A	N/A
7919713	12/2010	Al-Ali et al.	N/A	N/A

7937128	12/2010	Al-Ali	N/A	N/A
7937129	12/2010	Mason et al.	N/A	N/A
7941199	12/2010	Kiani	N/A	N/A
7957780	12/2010	Lamego et al.	N/A	N/A
7962188	12/2010	Kiani et al.	N/A	N/A
7976472	12/2010	Kiani	N/A	N/A
7990382	12/2010	Kiani	N/A	N/A
8008088	12/2010	Bellott et al.	N/A	N/A
RE42753	12/2010	Kiani- Azarbayjany et al.	N/A	N/A
8028701	12/2010	Al-Ali et al.	N/A	N/A
8048040	12/2010	Kiani	N/A	N/A
8050728	12/2010	Al-Ali et al.	N/A	N/A
RE43169	12/2011	Parker	N/A	N/A
8118620	12/2011	Al-Ali et al.	N/A	N/A
8130105	12/2011	Al-Ali et al.	N/A	N/A
8182443	12/2011	Kiani	N/A	N/A
8190223	12/2011	Al-Ali et al.	N/A	N/A
8203438	12/2011	Kiani et al.	N/A	N/A
8203704	12/2011	Merritt et al.	N/A	N/A
8219172	12/2011	Schurman et al.	N/A	N/A
8224411	12/2011	Al-Ali et al.	N/A	N/A
8229532	12/2011	Davis	N/A	N/A
8233955	12/2011	Al-Ali et al.	N/A	N/A
8255026	12/2011	Al-Ali	N/A	N/A
8265723	12/2011	McHale et al.	N/A	N/A
8274360	12/2011	Sampath et al.	N/A	N/A
8280473	12/2011	Al-Ali	N/A	N/A
8315683	12/2011	Al-Ali et al.	N/A	N/A
RE43860	12/2011	Parker	N/A	N/A
8346330	12/2012	Lamego	N/A	N/A
8353842	12/2012	Al-Ali et al.	N/A	N/A
8355766	12/2012	Macneish, III et al.	N/A	N/A
8374665	12/2012	Lamego	N/A	N/A
8388353	12/2012	Kiani et al.	N/A	N/A
8401602	12/2012	Kiani	N/A	N/A
8414499	12/2012	Al-Ali et al.	N/A	N/A
8418524	12/2012	Al-Ali	N/A	N/A
8428967	12/2012	Olsen et al.	N/A	N/A
8430817	12/2012	Al-Ali et al.	N/A	N/A
8437825	12/2012	Dalvi et al.	N/A	N/A
8455290	12/2012	Siskavich	N/A	N/A
8457707	12/2012	Kiani	N/A	N/A
8471713	12/2012	Poeze et al.	N/A	N/A
8473020	12/2012	Kiani et al.	N/A	N/A
8509867	12/2012	Workman et al.	N/A	N/A
8515509	12/2012	Bruinsma et al.	N/A	N/A
8523781	12/2012	Al-Ali	N/A	N/A
D692145	12/2012	Al-Ali et al.	N/A	N/A

8571617	12/2012	Reichgott et al.	N/A	N/A
8571618	12/2012	Lamego et al.	N/A	N/A
8571619	12/2012	Al-Ali et al.	N/A	N/A
8577431	12/2012	Lamego et al.	N/A	N/A
8584345	12/2012	Al-Ali et al.	N/A	N/A
8588880	12/2012	Abdul-Hafiz et al.	N/A	N/A
8630691	12/2013	Lamego et al.	N/A	N/A
8641631	12/2013	Sierra et al.	N/A	N/A
8652060	12/2013	Al-Ali	N/A	N/A
8666468	12/2013	Al-Ali	N/A	N/A
8670811	12/2013	O'Reilly	N/A	N/A
RE44823	12/2013	Parker	N/A	N/A
RE44875	12/2013	Kiani et al.	N/A	N/A
8688183	12/2013	Bruinsma et al.	N/A	N/A
8690799	12/2013	Telfort et al.	N/A	N/A
8702627	12/2013	Telfort et al.	N/A	N/A
8712494	12/2013	MacNeish et al.	N/A	N/A
8715206	12/2013	Telfort et al.	N/A	N/A
8723677	12/2013	Kiani	N/A	N/A
8740792	12/2013	Kiani et al.	N/A	N/A
8755535	12/2013	Telfort et al.	N/A	N/A
8755872	12/2013	Marinow	N/A	N/A
8764671	12/2013	Kiani	N/A	N/A
8768423	12/2013	Shakespeare et al.	N/A	N/A
8771204	12/2013	Telfort et al.	N/A	N/A
8781544	12/2013	Al-Ali et al.	N/A	N/A
8790268	12/2013	Al-Ali	N/A	N/A
8801613	12/2013	Al-Ali et al.	N/A	N/A
8821397	12/2013	Al-Ali et al.	N/A	N/A
8821415	12/2013	Al-Ali et al.	N/A	N/A
8830449	12/2013	Lamego et al.	N/A	N/A
8840549	12/2013	Al-Ali et al.	N/A	N/A
8852094	12/2013	Al-Ali et al.	N/A	N/A
8852994	12/2013	Wojtczuk et al.	N/A	N/A
8897847	12/2013	Al-Ali	N/A	N/A
8911377	12/2013	Al-Ali	N/A	N/A
8989831	12/2014	Al-Ali et al.	N/A	N/A
8998809	12/2014	Kiani	N/A	N/A
9066666	12/2014	Kiani	N/A	N/A
9066680	12/2014	Al-Ali et al.	N/A	N/A
9095316	12/2014	Welch et al.	N/A	N/A
9106038	12/2014	Telfort et al.	N/A	N/A
9107625	12/2014	Telfort et al.	N/A	N/A
9131881	12/2014	Diab et al.	N/A	N/A
9138180	12/2014	Coverston et al.	N/A	N/A
9153112	12/2014	Kiani et al.	N/A	N/A
9192329	12/2014	Al-Ali	N/A	N/A
9192351	12/2014	Telfort et al.	N/A	N/A
9195385	12/2014	Al-Ali et al.	N/A	N/A
9211095	12/2014	Al-Ali	N/A	N/A

9218454	12/2014	Kiani et al.	N/A	N/A
9245668	12/2015	Vo et al.	N/A	N/A
9267572	12/2015	Barker et al.	N/A	N/A
9277880	12/2015	Poeze et al.	N/A	N/A
9307928	12/2015	Al-Ali et al.	N/A	N/A
9323894	12/2015	Kiani	N/A	N/A
D755392	12/2015	Hwang et al.	N/A	N/A
9326712	12/2015	Kiani	N/A	N/A
9392945	12/2015	Al-Ali et al.	N/A	N/A
9408542	12/2015	Kinast et al.	N/A	N/A
9436645	12/2015	Al-Ali et al.	N/A	N/A
9445759	12/2015	Lamego et al.	N/A	N/A
9474474	12/2015	Lamego et al.	N/A	N/A
9480435	12/2015	Olsen	N/A	N/A
9510779	12/2015	Poeze et al.	N/A	N/A
9517024	12/2015	Kiani et al.	N/A	N/A
9532722	12/2016	Lamego et al.	N/A	N/A
9560996	12/2016	Kiani	N/A	N/A
9579039	12/2016	Jansen et al.	N/A	N/A
9622692	12/2016	Lamego et al.	N/A	N/A
D788312	12/2016	Al-Ali et al.	N/A	N/A
9649054	12/2016	Lamego et al.	N/A	N/A
9697928	12/2016	Al-Ali et al.	N/A	N/A
9717458	12/2016	Lamego et al.	N/A	N/A
9724016	12/2016	Al-Ali et al.	N/A	N/A
9724024	12/2016	Al-Ali	N/A	N/A
9724025	12/2016	Kiani et al.	N/A	N/A
9749232	12/2016	Sampath et al.	N/A	N/A
9750442	12/2016	Olsen	N/A	N/A
9750461	12/2016	Telfort	N/A	N/A
9775545	12/2016	Al-Ali et al.	N/A	N/A
9778079	12/2016	Al-Ali et al.	N/A	N/A
9782077	12/2016	Lamego et al.	N/A	N/A
9787568	12/2016	Lamego et al.	N/A	N/A
9808188	12/2016	Perea et al.	N/A	N/A
9839379	12/2016	Al-Ali et al.	N/A	N/A
9839381	12/2016	Weber et al.	N/A	N/A
9847749	12/2016	Kiani et al.	N/A	N/A
9848800	12/2016	Lee et al.	N/A	N/A
9861298	12/2017	Eckerbom et al.	N/A	N/A
9861305	12/2017	Weber et al.	N/A	N/A
9877650	12/2017	Muhsin et al.	N/A	N/A
9891079	12/2017	Dalvi	N/A	N/A
9924897	12/2017	Abdul-Hafiz	N/A	N/A
9936917	12/2017	Poeze et al.	N/A	N/A
9955937	12/2017	Telfort	N/A	N/A
9965946	12/2017	Al-Ali et al.	N/A	N/A
D820865	12/2017	Muhsin et al.	N/A	N/A
9986952	12/2017	Dalvi et al.	N/A	N/A
D822215	12/2017	Al-Ali et al.	N/A	N/A

D822216	12/2017	Barker et al.	N/A	N/A
10010276	12/2017	Al-Ali et al.	N/A	N/A
10086138	12/2017	Novak, Jr.	N/A	N/A
10111591	12/2017	Dyell et al.	N/A	N/A
D833624	12/2017	DeJong et al.	N/A	N/A
10123729	12/2017	Dyell et al.	N/A	N/A
D835282	12/2017	Barker et al.	N/A	N/A
D835283	12/2017	Barker et al.	N/A	N/A
D835284	12/2017	Barker et al.	N/A	N/A
D835285	12/2017	Barker et al.	N/A	N/A
10149616	12/2017	Al-Ali et al.	N/A	N/A
10154815	12/2017	Al-Ali et al.	N/A	N/A
10159412	12/2017	Lamego et al.	N/A	N/A
10188348	12/2018	Al-Ali et al.	N/A	N/A
RE47218	12/2018	Al-Ali	N/A	N/A
RE47244	12/2018	Kiani et al.	N/A	N/A
RE47249	12/2018	Kiani et al.	N/A	N/A
10205291	12/2018	Scruggs et al.	N/A	N/A
10226187	12/2018	Al-Ali et al.	N/A	N/A
10231657	12/2018	Al-Ali et al.	N/A	N/A
10231670	12/2018	Blank et al.	N/A	N/A
RE47353	12/2018	Kiani et al.	N/A	N/A
10279247	12/2018	Kiani	N/A	N/A
10292664	12/2018	Al-Ali	N/A	N/A
10299720	12/2018	Brown et al.	N/A	N/A
10327337	12/2018	Schmidt et al.	N/A	N/A
10327713	12/2018	Barker et al.	N/A	N/A
10332630	12/2018	Al-Ali	N/A	N/A
10383520	12/2018	Wojtczuk et al.	N/A	N/A
10383527	12/2018	Al-Ali	N/A	N/A
10388120	12/2018	Muhsin et al.	N/A	N/A
D864120	12/2018	Forrest et al.	N/A	N/A
10441181	12/2018	Telfort et al.	N/A	N/A
10441196	12/2018	Eckerbom et al.	N/A	N/A
10448844	12/2018	Al-Ali et al.	N/A	N/A
10448871	12/2018	Al-Ali et al.	N/A	N/A
10456038	12/2018	Lamego et al.	N/A	N/A
10463340	12/2018	Telfort et al.	N/A	N/A
10471159	12/2018	Lapotko et al.	N/A	N/A
10505311	12/2018	Al-Ali et al.	N/A	N/A
10524738	12/2019	Olsen	N/A	N/A
10532174	12/2019	Al-Ali	N/A	N/A
10537285	12/2019	Shreim et al.	N/A	N/A
10542903	12/2019	Al-Ali et al.	N/A	N/A
10555678	12/2019	Dalvi et al.	N/A	N/A
10568553	12/2019	O'Neil et al.	N/A	N/A
RE47882	12/2019	Al-Ali	N/A	N/A
10608817	12/2019	Haider et al.	N/A	N/A
D880477	12/2019	Forrest et al.	N/A	N/A
10617302	12/2019	Al-Ali et al.	N/A	N/A

10617335	12/2019	Al-Ali et al.	N/A	N/A
10637181	12/2019	Al-Ali et al.	N/A	N/A
D886849	12/2019	Muhsin et al.	N/A	N/A
D887548	12/2019	Abdul-Hafiz et al.	N/A	N/A
D887549	12/2019	Abdul-Hafiz et al.	N/A	N/A
10667764	12/2019	Ahmed et al.	N/A	N/A
D890708	12/2019	Forrest et al.	N/A	N/A
10721785	12/2019	Al-Ali	N/A	N/A
10736518	12/2019	Al-Ali et al.	N/A	N/A
10750984	12/2019	Pauley et al.	N/A	N/A
D897098	12/2019	Al-Ali	N/A	N/A
10779098	12/2019	Iswanto et al.	N/A	N/A
10827961	12/2019	Iyengar et al.	N/A	N/A
10828007	12/2019	Telfort et al.	N/A	N/A
10832818	12/2019	Muhsin et al.	N/A	N/A
10849554	12/2019	Shreim et al.	N/A	N/A
10856750	12/2019	Indorf	N/A	N/A
D906970	12/2020	Forrest et al.	N/A	N/A
D908213	12/2020	Abdul-Hafiz et al.	N/A	N/A
10918281	12/2020	Al-Ali et al.	N/A	N/A
10932705	12/2020	Muhsin et al.	N/A	N/A
10932729	12/2020	Kiani et al.	N/A	N/A
10939878	12/2020	Kiani et al.	N/A	N/A
10952641	12/2020	Eckerbom et al.	N/A	N/A
10956950	12/2020	Al-Ali et al.	N/A	N/A
D916135	12/2020	Indorf et al.	N/A	N/A
D917046	12/2020	Abdul-Hafiz et al.	N/A	N/A
D917550	12/2020	Indorf et al.	N/A	N/A
D917564	12/2020	Indorf et al.	N/A	N/A
D917704	12/2020	Al-Ali et al.	N/A	N/A
10987066	12/2020	Chandran et al.	N/A	N/A
10991135	12/2020	Al-Ali et al.	N/A	N/A
D919094	12/2020	Al-Ali et al.	N/A	N/A
D919100	12/2020	Al-Ali et al.	N/A	N/A
11006867	12/2020	Al-Ali	N/A	N/A
D921202	12/2020	Al-Ali et al.	N/A	N/A
11024064	12/2020	Muhsin et al.	N/A	N/A
11026604	12/2020	Chen et al.	N/A	N/A
D925597	12/2020	Chandran et al.	N/A	N/A
D927699	12/2020	Al-Ali et al.	N/A	N/A
11076777	12/2020	Lee et al.	N/A	N/A
11114188	12/2020	Poeze et al.	N/A	N/A
D933232	12/2020	Al-Ali et al.	N/A	N/A
D933233	12/2020	Al-Ali et al.	N/A	N/A
D933234	12/2020	Al-Ali et al.	N/A	N/A
11145408	12/2020	Sampath et al.	N/A	N/A
11147518	12/2020	Al-Ali et al.	N/A	N/A
11185262	12/2020	Al-Ali et al.	N/A	N/A
11191484	12/2020	Kiani et al.	N/A	N/A
D946596	12/2021	Ahmed	N/A	N/A

D946597	12/2021	Ahmed	N/A	N/A
D946598	12/2021	Ahmed	N/A	N/A
D946617	12/2021	Ahmed	N/A	N/A
11272839	12/2021	Al-Ali et al.	N/A	N/A
11289199	12/2021	Al-Ali	N/A	N/A
RE49034	12/2021	Al-Ali	N/A	N/A
11298021	12/2021	Muhsin et al.	N/A	N/A
D950580	12/2021	Ahmed	N/A	N/A
D950599	12/2021	Ahmed	N/A	N/A
D950738	12/2021	Al-Ali et al.	N/A	N/A
D957648	12/2021	Al-Ali	N/A	N/A
11389093	12/2021	Triman et al.	N/A	N/A
11406286	12/2021	Al-Ali et al.	N/A	N/A
11417426	12/2021	Muhsin et al.	N/A	N/A
11439329	12/2021	Lamego	N/A	N/A
11445948	12/2021	Scruggs et al.	N/A	N/A
D965789	12/2021	Al-Ali et al.	N/A	N/A
D967433	12/2021	Al-Ali et al.	N/A	N/A
11464410	12/2021	Muhsin	N/A	N/A
11504058	12/2021	Sharma et al.	N/A	N/A
11504066	12/2021	Dalvi et al.	N/A	N/A
D971933	12/2021	Ahmed	N/A	N/A
D973072	12/2021	Ahmed	N/A	N/A
D973685	12/2021	Ahmed	N/A	N/A
D973686	12/2021	Ahmed	N/A	N/A
D974193	12/2022	Forrest et al.	N/A	N/A
11564593	12/2022	Eckerbom et al.	N/A	N/A
D979516	12/2022	Al-Ali et al.	N/A	N/A
D980091	12/2022	Forrest et al.	N/A	N/A
11596363	12/2022	Lamego	N/A	N/A
2001/0034477	12/2000	Mansfield et al.	N/A	N/A
2001/0039483	12/2000	Brand et al.	N/A	N/A
2002/0010401	12/2001	Bushmakina et al.	N/A	N/A
2002/0058864	12/2001	Mansfield et al.	N/A	N/A
2002/0077606	12/2001	Trotta	N/A	N/A
2002/0133080	12/2001	Apruzzese et al.	N/A	N/A
2002/0138093	12/2001	Song	606/191	A61M 60/857
2003/0013975	12/2002	Kiani	N/A	N/A
2003/0018243	12/2002	Gerhardt et al.	N/A	N/A
2003/0144582	12/2002	Cohen et al.	N/A	N/A
2003/0156288	12/2002	Barnum et al.	N/A	N/A
2003/0191405	12/2002	Rich et al.	N/A	N/A
2003/0212312	12/2002	Coffin, IV et al.	N/A	N/A
2004/0029467	12/2003	Lacroix	N/A	N/A
2004/0106163	12/2003	Workman, Jr. et al.	N/A	N/A
2005/0055276	12/2004	Kiani et al.	N/A	N/A
2005/0161042	12/2004	Fudge et al.	N/A	N/A
2005/0234317	12/2004	Kiani	N/A	N/A

2006/0014059	12/2005	Wood	N/A	N/A
2006/0073719	12/2005	Kiani	N/A	N/A
2006/0086254	12/2005	Fudge et al.	N/A	N/A
2006/0189871	12/2005	Al-Ali et al.	N/A	N/A
2006/0200110	12/2005	Lentz et al.	N/A	N/A
2007/0073116	12/2006	Kiani et al.	N/A	N/A
2007/0180140	12/2006	Welch et al.	N/A	N/A
2007/0244377	12/2006	Cozad et al.	N/A	N/A
2008/0034964	12/2007	Schmidt et al.	N/A	N/A
2008/0064965	12/2007	Jay et al.	N/A	N/A
2008/0094228	12/2007	Welch et al.	N/A	N/A
2008/0221418	12/2007	Al-Ali et al.	N/A	N/A
2009/0036759	12/2008	Ault et al.	N/A	N/A
2009/0093687	12/2008	Telfort et al.	N/A	N/A
2009/0095926	12/2008	MacNeish, III	N/A	N/A
2009/0247984	12/2008	Lamego et al.	N/A	N/A
2010/0004518	12/2009	Vo et al.	N/A	N/A
2010/0030040	12/2009	Poeze et al.	N/A	N/A
2010/0099964	12/2009	O'Reilly et al.	N/A	N/A
2010/0174239	12/2009	Yodfat et al.	N/A	N/A
2010/0234718	12/2009	Sampath et al.	N/A	N/A
2010/0270257	12/2009	Wachman et al.	N/A	N/A
2011/0028806	12/2010	Merritt et al.	N/A	N/A
2011/0028809	12/2010	Goodman	N/A	N/A
2011/0040197	12/2010	Welch et al.	N/A	N/A
2011/0082711	12/2010	Poeze et al.	N/A	N/A
2011/0087081	12/2010	Kiani et al.	N/A	N/A
2011/0118561	12/2010	Tari et al.	N/A	N/A
2011/0137297	12/2010	Kiani et al.	N/A	N/A
2011/0172498	12/2010	Olsen et al.	N/A	N/A
2011/0230733	12/2010	Al-Ali	N/A	N/A
2011/0237969	12/2010	Eckerbom et al.	N/A	N/A
2012/0123231	12/2011	O'Reilly	N/A	N/A
2012/0165629	12/2011	Merritt et al.	N/A	N/A
2012/0209084	12/2011	Olsen et al.	N/A	N/A
2012/0226117	12/2011	Lamego et al.	N/A	N/A
2012/0283524	12/2011	Kiani et al.	N/A	N/A
2013/0023775	12/2012	Lamego et al.	N/A	N/A
2013/0041591	12/2012	Lamego	N/A	N/A
2013/0060147	12/2012	Welch et al.	N/A	N/A
2013/0096405	12/2012	Garfio	N/A	N/A
2013/0296672	12/2012	O'Neil et al.	N/A	N/A
2013/0345921	12/2012	Al-Ali et al.	N/A	N/A
2014/0166076	12/2013	Kiani et al.	N/A	N/A
2014/0180160	12/2013	Brown et al.	N/A	N/A
2014/0187973	12/2013	Brown et al.	N/A	N/A
2014/0275871	12/2013	Lamego et al.	N/A	N/A
2014/0275872	12/2013	Merritt et al.	N/A	N/A
2014/0316217	12/2013	Purdon et al.	N/A	N/A
2014/0316218	12/2013	Purdon et al.	N/A	N/A

2014/0323897	12/2013	Brown et al.	N/A	N/A
2014/0323898	12/2013	Purdon et al.	N/A	N/A
2015/0005600	12/2014	Blank et al.	N/A	N/A
2015/0011907	12/2014	Purdon et al.	N/A	N/A
2015/0073241	12/2014	Lamego	N/A	N/A
2015/0080754	12/2014	Purdon et al.	N/A	N/A
2015/0099950	12/2014	Al-Ali et al.	N/A	N/A
2016/0196388	12/2015	Lamego	N/A	N/A
2016/0367173	12/2015	Dalvi et al.	N/A	N/A
2017/0024748	12/2016	Haider	N/A	N/A
2017/0042488	12/2016	Muhsin	N/A	N/A
2017/0173632	12/2016	Al-Ali	N/A	N/A
2017/0251974	12/2016	Shreim et al.	N/A	N/A
2017/0311891	12/2016	Kiani et al.	N/A	N/A
2018/0103874	12/2017	Lee et al.	N/A	N/A
2018/0242926	12/2017	Muhsin et al.	N/A	N/A
2018/0247353	12/2017	Al-Ali et al.	N/A	N/A
2018/0247712	12/2017	Muhsin et al.	N/A	N/A
2018/0256087	12/2017	Al-Ali et al.	N/A	N/A
2018/0296161	12/2017	Shreim et al.	N/A	N/A
2018/0300919	12/2017	Muhsin et al.	N/A	N/A
2018/0310822	12/2017	Indorf et al.	N/A	N/A
2018/0310823	12/2017	Al-Ali et al.	N/A	N/A
2018/0317826	12/2017	Muhsin et al.	N/A	N/A
2019/0015023	12/2018	Monfre	N/A	N/A
2019/0117070	12/2018	Muhsin et al.	N/A	N/A
2019/0200941	12/2018	Chandran et al.	N/A	N/A
2019/0239787	12/2018	Pauley et al.	N/A	N/A
2019/0320906	12/2018	Olsen	N/A	N/A
2019/0374139	12/2018	Kiani et al.	N/A	N/A
2019/0374173	12/2018	Kiani et al.	N/A	N/A
2019/0374713	12/2018	Kiani et al.	N/A	N/A
2020/0060869	12/2019	Telfort et al.	N/A	N/A
2020/0111552	12/2019	Ahmed	N/A	N/A
2020/0113435	12/2019	Muhsin	N/A	N/A
2020/0113488	12/2019	Al-Ali et al.	N/A	N/A
2020/0113496	12/2019	Scruggs et al.	N/A	N/A
2020/0113497	12/2019	Triman et al.	N/A	N/A
2020/0113520	12/2019	Abdul-Hafiz et al.	N/A	N/A
2020/0138288	12/2019	Al-Ali et al.	N/A	N/A
2020/0138368	12/2019	Kiani et al.	N/A	N/A
2020/0163597	12/2019	Dalvi et al.	N/A	N/A
2020/0196877	12/2019	Vo et al.	N/A	N/A
2020/0253474	12/2019	Muhsin et al.	N/A	N/A
2020/0253544	12/2019	Belur Nagaraj et al.	N/A	N/A
2020/0275841	12/2019	Telfort et al.	N/A	N/A
2020/0288983	12/2019	Telfort et al.	N/A	N/A
2020/0321793	12/2019	Al-Ali et al.	N/A	N/A
2020/0329983	12/2019	Al-Ali et al.	N/A	N/A

2020/0329984	12/2019	Al-Ali et al.	N/A	N/A
2020/0329993	12/2019	Al-Ali et al.	N/A	N/A
2020/0330037	12/2019	Al-Ali et al.	N/A	N/A
2021/0022628	12/2020	Telfort et al.	N/A	N/A
2021/0104173	12/2020	Pauley et al.	N/A	N/A
2021/0113121	12/2020	Diab et al.	N/A	N/A
2021/0117525	12/2020	Kiani et al.	N/A	N/A
2021/0118581	12/2020	Kiani et al.	N/A	N/A
2021/0121582	12/2020	Krishnamani et al.	N/A	N/A
2021/0161465	12/2020	Barker et al.	N/A	N/A
2021/0236729	12/2020	Kiani et al.	N/A	N/A
2021/0256267	12/2020	Ranasinghe et al.	N/A	N/A
2021/0256835	12/2020	Ranasinghe et al.	N/A	N/A
2021/0275101	12/2020	Vo et al.	N/A	N/A
2021/0290060	12/2020	Ahmed	N/A	N/A
2021/0290072	12/2020	Forrest	N/A	N/A
2021/0290080	12/2020	Ahmed	N/A	N/A
2021/0290120	12/2020	Al-Ali	N/A	N/A
2021/0290177	12/2020	Novak, Jr.	N/A	N/A
2021/0290184	12/2020	Ahmed	N/A	N/A
2021/0296008	12/2020	Novak, Jr.	N/A	N/A
2021/0330228	12/2020	Olsen et al.	N/A	N/A
2021/0386382	12/2020	Olsen et al.	N/A	N/A
2021/0402110	12/2020	Pauley et al.	N/A	N/A
2022/0026355	12/2021	Normand et al.	N/A	N/A
2022/0039707	12/2021	Sharma et al.	N/A	N/A
2022/0053892	12/2021	Al-Ali et al.	N/A	N/A
2022/0071562	12/2021	Kiani	N/A	N/A
2022/0096603	12/2021	Kiani et al.	N/A	N/A
2022/0151521	12/2021	Krishnamani et al.	N/A	N/A
2022/0218244	12/2021	Kiani et al.	N/A	N/A
2022/0287574	12/2021	Telfort et al.	N/A	N/A
2022/0296161	12/2021	Al-Ali et al.	N/A	N/A
2022/0361819	12/2021	Al-Ali et al.	N/A	N/A
2022/0379059	12/2021	Yu et al.	N/A	N/A
2022/0392610	12/2021	Kiani et al.	N/A	N/A
2023/0028745	12/2022	Al-Ali	N/A	N/A
2023/0038389	12/2022	Vo	N/A	N/A
2023/0045647	12/2022	Vo	N/A	N/A
2023/0058052	12/2022	Al-Ali	N/A	N/A
2023/0058342	12/2022	Kiani	N/A	N/A
2023/0069789	12/2022	Koo et al.	N/A	N/A
2023/0087671	12/2022	Telfort et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2175208	12/1985	GB	N/A
WO 98/046277	12/1997	WO	N/A
WO 2005/072297	12/2004	WO	N/A
WO 2006/120683	12/2005	WO	N/A

OTHER PUBLICATIONS

US 2022/0192529 A1, 06/2022, Al-Ali et al. (withdrawn) cited by applicant

740SELECT Capnography: Covidien Microstream MicroPod Brochure (publication date not provided) (Exhibit 1014). cited by applicant

Arkema Brochure, in Medical Applications—Advantage of Using Pebax (publication date not provided) (Exhibit 1011). cited by applicant

Barrer, Source: <http://en.wikipedia.org/wiki/Barrer?oldid=646860514>. Contributors: Calair, Wtmitchell, MZMcBride, Retired username, SmackBot, Bluebot, Due Freq, Flscholten, ydebot, Zginder, MarchBot, Leyo, Reedy Bot, Keepday, Steven J. Anderson, MacIgwmbot, Addbot, Colapeninsula, Barrercorrector and Anonymous: 10. As printed on Jun. 2, 2015. cited by applicant
Corrected Petition for Inter Parties Review of U.S. Pat. No. 9,861,298 dated Sep. 6, 2019. cited by applicant

Credit Suisse, Report on PEEK Pricing, dated May 13, 2015 (Exhibit 1013). cited by applicant

Curriculum Vitae, Len Czuba [Exhibit 2003]. cited by applicant

Decision Granting Institution of Inter Partes Review date Mar. 16, 2019. cited by applicant

Declaration of Craig Sunada in *Perma Pure, LLC V. Masimo Corporation* dated Aug. 24, 2020 in 20 pages. [Exhibit 1028]. cited by applicant

Declaration of Len Czuba [Exhibit 2002]. cited by applicant

Declaration of Petitioner's Expert, Zane Frund, Ph.D., dated Sep. 6, 2019 (Exhibit 1002). cited by applicant

Deposition of Leonard F. Czuba in *Perma Pure, LLC V. Masimo Corporation* dated Aug. 13, 2020 in 113 pages. [Exhibit 1027]. cited by applicant

E. Peled et al., A Direct Methanol Fuel Cell Based on a Novel Low-Cost Nanoporous Proton-Conducting Membrane, dated Jun. 26, 2000 (Exhibit 1025). cited by applicant

European Patent Office Third Party Observations dated Jun. 8, 2015, Application No. 09813309.3, Patent No. 2326246, WO 2010/030226. cited by applicant

European Patent Office Third Party Observations, dated Sep. 28, 2012; Appln. No. 09813309.3-2319 / 2326246 PCT/SE2009051012. cited by applicant

Examiners Answer Brief, dated Mar. 28, 2016 (Exhibit 1023). cited by applicant

Excerpts from Mulder, Marcel, Basic Principles of Membrane Technology, 1996 (2nd ed.) [Exhibit 2010]. cited by applicant

File History, U.S. Appl. No. 13/063,648 [Exhibit 2001]. cited by applicant

Flyer; "Our Materials Science Meets Your Medical Applications", Arkema 2009; 6 pages. cited by applicant

Freeman et al., Gas and Liquid Separations Using Membranes: An Overview, 2004 American Chemical Society (Exhibit 1021). cited by applicant

International Publication No. WO 2005/072297, RIC Investments, LLC, published Aug. 11, 2005 [Exhibit 2005] [Exhibit 2015]. cited by applicant

International Search Report: PCT/SE2009/051012; mailed Dec. 21, 2009. cited by applicant

Jansen, Nanotubes Vapour Removal, Attentie, Periodiek der S.V.A.T. Astante, Oct. 2007 (Exhibit 1017). cited by applicant

Ma et al., Carbon Dioxide Permeability of Proton Exchange Membranes for Fuel Cells, Solid State Ionics 176 (2005) p. 2923-2927 (Exhibit 1015). cited by applicant

Marcq, J. et al., "Abatement of CO₂ Emissions by Means of Membranes—Characterization of Industrial PEBAX™ Films", Environment Protection Engineering, 2005, vol. 31 (3-4), pp. 13-22. cited by applicant

Moilanen, et al., Water Dynamics and Proton Transfer in Nafion Fuel Cell Membranes, Langmuir 2008, 24(8), 3690-3698. Published Apr. 15, 2008. cited by applicant

Nguyen, et al., Pervaporation, a Novel Technique for the Measurement of Vapor Transmission Rate of Highly Permeable Films, Polymer Testing 29 (2001) p. 901-911 (Exhibit 1007). cited by applicant

Patent Owner Preliminary Response dated Dec. 19, 2019. cited by applicant

Patent Owner Submission of Mandatory Notice Information Pursuant to 37 C.F.R. 42.8(a) dated Sep. 27, 2019. cited by applicant

PEBAX Invioce, dated Jul. 18, 2019 (Exhibit 1026). cited by applicant

Pebax®, Applications Areas, Arkema, Jun. 2000, 11 pages. cited by applicant

Pebax®, Breathable Film, Arkema, Jun. 2007, 2 pages. cited by applicant

Pebax®, in Medical Applications, Arkema, Jun. 2007, 2 pages. cited by applicant

Pebax®, Product Range Overview, Arkema, Jun. 2007, 2 pages. cited by applicant

Perma Pure LLC, Frequently Asked Questions, <https://web.archive.org/web/20060324193926/http://www.permapure.com/FAQs.htm> [Exhibit 2007]. cited by applicant

Perma Pure, LLC, Frequently Asked Questions, <https://www.permapure.com/support/faqs/> [Exhibit 2008]. cited by applicant

Perma Pure, LLC, Press Release, “Fluid Technology Division—Keeping Water Vapor From Breath Gas Analyzers,” Feb. 18, 2005. [Exhibit 2012]. cited by applicant

Petition for Inter Parties Review of U.S. Pat. No. 9,861,298 dated Sep. 6, 2019. cited by applicant

Petitioner's Opening Appeal Brief, dated Nov. 30, 2015(Exhibit 1022). cited by applicant

Petitioner's Reply Brief to Examiner's Answer, dated Jun. 6, 2016 (Exhibit 1024). cited by applicant

Portion of File History of European Application No. 098133093.-1660, Jun. 8, 2015 (Exhibit 1018). cited by applicant

Potreck, J., et al., “Mixed Water Vapor/Gas Transport Through the rubbery polymer PEBAX® 1074,” Journal of Membrane Science, 338 (2009). [Exhibit 2011]. cited by applicant

Prosecution History of U.S. Pat. No. 9,861,298, patent issued on Jan. 9, 2018 (Exhibit 1009). cited by applicant

Record of Oral Hearing Held on Dec. 15, 2020 for IPR2019-01583, dated Feb. 12, 2021. cited by applicant

Second Portion of File History of European Application No. 098133093.-1660, Jun. 8, 2015 (Exhibit 1019). cited by applicant

Shamu, A., et al., “Mass transfer studies on the dehydration of supercritical carbon dioxide using dense polymeric membranes,” Separation and Purification Technology, 209 (2019). [Exhibit 2013]. cited by applicant

Sijbesma et al., “Flue Gas Dehydration Using Polymer Membranes”, Journal of Membrane Science, vol. 313, 2008, pp. 263-276. cited by applicant

Sijbesma et al., Flue Gas Dehydration Using Polymer Membranes, J. Membrane Sci 313 (2008) p. 263-276 (Exhibit 1004). cited by applicant

Španěl, P., et al., “On-line measurement of the absolute humidity of air, breath and liquid headspace samples by selected ion flow tube mass spectrometry,” Rapid Communications in Mass Spectrometry, 15 (2001). [Exhibit 2009]. cited by applicant

Sridhar et al., “Development of Crosslinked Polyether-block-amide membrane for CO₂/CH₄ Separation,” Colloids and Surfaces A: Physiocochem, Eng. Aspects 297, 2007, pp. 267-274 (Exhibit 1016). cited by applicant

Supplementary European Search Report completed Nov. 30, 2012; Appln. No. EP 09 81 3309. cited by applicant

Third Portion of File History of European Application No. 098133093.-1660, Jun. 8, 2015 (Exhibit 1020). cited by applicant

U.S. Pat. No. 6,779,522, Smith et al., Issued Aug. 24, 2004 [Exhibit 2006]. cited by applicant

U.S. Pat. No. 6,783,573, Richardson, Issued Aug. 31, 2004 (Exhibit 1012 & Exhibit 2004). cited by

applicant

U.S. Pat. No. 6,783,573, Richardson, Issued Aug. 31, 2004 [Exhibit 2004] [Exhibit 2014]. cited by applicant

U.S. Pat. No. 8,053,030, Gilman, Issued Nov. 8, 2011 (Exhibit 1006). cited by applicant

U.S. Pat. No. 9,861,298, Eckerbom et al., Issued Jan. 9, 2018 (Exhibit 1001). cited by applicant

U.S. Pat. No. 95,042,500, Norlien et al., Issued Aug. 27, 1991 (Exhibit 1003). cited by applicant

US Publication 2005/0171496, Guldeldt et al., Published Aug. 4, 2005 (Exhibit 1005). cited by applicant

US Publication 2009/0088656, Levitsky et al., Published Apr. 2, 2009 (Exhibit 1008). cited by applicant

US Publication No. 2003/0191405, Rich et al., Issued Oct. 9, 2003 [Exhibit 2016]. cited by applicant

USPTO, Decision on Appeal 2016-006238, Exhibit 1010, filed on Sep. 6, 2019 in IPR2019-01583 dated Sep. 18, 2017 (Exhibit 1010). cited by applicant

V.I. Bondar, et al; "Gas Sorption and Characterization of Poly(ether-b-amide) Segmented Block Copolymers", Journal of Polymer Science: Part B: Polymer Physics, vol. 37, pp. 2463-2475, Apr. 26, 1999. cited by applicant

Wikipedia.org: PEBA and PEG. cited by applicant

Yuri Yampolskii, et al; "Membrane Gas Separation", John Wiley & Sons, Ltd, 2010; pp. 257-263. cited by applicant

Primary Examiner: Agahi; Puya

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a gas sampling line having a channel for conducting respiratory gases from a patient respiratory interface to a gas monitor, and to a gas analysis system comprising such a gas sampling line. Furthermore, the invention relates to the use of a tube for sampling of respiratory gases, and to a method for sampling of respiratory gases.

BACKGROUND ART

(2) In respiratory care, it is often desirable to analyse and monitor the gas composition of a patient's exhaled and/or inhaled breathing gases. For instance, measurement of respiratory CO₂, O₂, N₂O and anesthetic agents, such as halothane, isoflurane, enflurane, sevoflurane or desflurane, is useful in the care of critically ill patients undergoing anesthesia or mechanical ventilation. Typically, the gas concentrations of the patient's breathing gases are monitored by transferring a portion of the breathing gases through a sampling line to a suitable gas sensor or gas monitor.

(3) The patient's exhaled breathing gases are usually saturated with moisture at body temperature. Thus, water naturally condenses when the gas sample is cooled to room temperature when passed through the sampling line. Collected condensate, together with secretion, bacteria or other contaminants possibly present in the breathing gases, may result in inaccurate readings at the sensor or even adversely affect a delicate gas monitor.

(4) The accuracy of the gas concentrations obtained from a respiratory gas monitor also depends on the ability of the analyser system to direct the gas sample from the patient, through the tube of a sampling line to the gas sensor, without distorting the gas sample flow. One cause of distortion of

the gas sample flow may be the adsorption on and/or absorption in the tube material of one of more of the components of the gas sample. Additionally, any physical obstacles in the gas sample line, such as valves or material seams, may distort the gas sample flow. Distortion of the gas sample flow, regardless of cause, can degrade rise time of the measured waveform making accurate analysis, especially at higher breath rates, difficult or impossible.

(5) In order to protect a respiratory gas monitor from water and other contaminants, prior art gas sampling systems employs various means to separate liquids, bacteria etc. from the gas sample flow. It has for instance been known to include in the sampling line a water trap or another moisture separation means between the patient and the gas sensor. The challenge, however, is to design such a water trap or moisture separation means that achieves sufficient efficiency and capacity without distorting the gas sample flow.

(6) U.S. Pat. No. 6,783,573 is directed to a gas sampling system for conducting respiratory gases. A gas sampling tube is configured to conduct the respiratory gases from a patient to a gas monitoring device. An output connector couples the gas sampling tube to the monitor. An output dryer tube is coupled between the gas sampling tube and the output connector. The output dryer tube is characterized by a tube length and a relative moisture removal efficiency. The relative moisture removal efficiency is dependent on the tube length. The tube length is selected to limit the moisture content of the respiratory gases being directed into the respiratory gas monitor to a predetermined level. The output dryer tube may be comprised of Nafion® or may be implemented using microporous filters or molecular sieves. An optional input dryer may be implemented using the same materials used to implement the output dryer.

(7) WO 2005/072297 is directed to a liquid absorbing filter assembly and system using the same. It is disclosed a filter assembly for use in a sidestream gas sampling assembly. The filter assembly includes a hydrophilic liner lining the inner perimeter of a housing for wicking moisture from the gases to be monitored prior to the gases reaching a sensing mechanism. It is contemplated that the housing can be formed from an absorbent material or a gas drying material, such as Nafion®.

(8) However, there is a need for improvement of prior art solutions for respiratory gas analysis in respect of, e.g., low distortion, long lasting moisture and/or water removal, or low cost.

SUMMARY OF THE INVENTION

(9) An object of the present invention is to provide means for sampling of respiratory gases from a patient while protecting a gas monitor from moisture and/or water as well as providing a reading of high accuracy of gas components, including air gases, such as CO₂, N₂O and anesthetic agents. Thus, it is an object of the invention to allow for a signal having a low distortion, particularly in a low flow gas sample and/or a gas sample from a patient having a high breath rate.

(10) Another object of the present invention is to provide means for sampling of respiratory gases from a patient without employment of a conventional water trap. It is thus an object of the invention to provide a simple and cheap solution in comparison to conventional means comprising such water trap as well as to provide prolonged operating times in comparison to those of conventional means.

(11) A further object of the present invention is to provide means of low material and production cost fulfilling the mentioned aspects of moisture and/or water removal as well as accurate gas component reading.

(12) Another object of the present invention is to improve sidestream measurements of respiratory gases, i.e. when in order to influence a patient as little as possible a small fraction only of the inhaled and/or exhaled respiratory gas is diverted for measurement from the main flow of respiratory gas between a patient and, e.g. a breather apparatus.

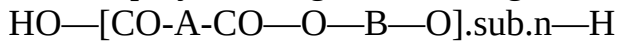
(13) Other objects or advantages of the invention should be apparent to a person skilled in the art after having read the description below.

(14) In a first aspect of the invention, there is provided a gas sampling line having a channel for conducting respiratory gases from a patient respiratory interface to a gas monitor, the gas sampling

line comprising a patient respiratory interface connector adapted to couple the gas sampling line to a patient respiratory interface; a gas sampling tube adapted to conduct respiratory gases; and a gas monitor connector adapted to couple the gas sampling line to a gas monitor, wherein the gas sampling tube is comprised of a polyether block amide material, the polyether segments of which comprise polyethyleneoxide.

(15) Thus, it has been surprisingly found that when passing a sample of respiratory gases through a gas sampling tube comprised of a polyether block amide material, the polyether segments of which comprise polyethyleneoxide, moisture and condensed water present in the gas sample permeate through the tube material to provide a dried gas sample, while components (such as CO₂ or anesthetic agents) of the gas sample passes the tube portion essentially undistorted (i.e. without being absorbed in or adsorbed to the tube material) to allow for an accurate reading at the gas monitor.

(16) Polyether block amides (PEBAs) are thermoplastic elastomers well adapted for applications such as molded or extruded articles, films etc. They are block copolymers obtained by polycondensation of a carboxylic acid polyamide with an alcohol termination polyether. A polyether block amide elastomer consists of a regular linear chain of rigid polyamide segments and flexible polyether segments having the following general formula



where A represents the polyamide segment and B represent the polyether segment. Polyamide is a polymer comprising the characteristic amide group
—NH—CO—

in the repeating units of the polymer chain. In the present invention, the polyether segments of the polyether block amide material comprise polyethyleneoxide, i.e. a polymer of the formula
 $\text{HO}—[\text{CH}_2—\text{CH}_2—\text{O}].\text{sub.n}-\text{H}$.

(17) The polyamide segments of the polyether block amide material may comprise polyamide-12, polyamide-11 or polyamide-12.12, preferably polyamide-12. The nomenclature of the polyamides corresponds to an internationally recognised system, where a number indicates the number of carbon atoms in a compound used to prepare the polyamide. If only one number is given, it means that the polyamide is derived from a aminocarboxylic acid having that number of carbon atoms. If two numbers are given, the first number indicates the number of carbon atoms of a starting diamine and the last number indicates the number of carbon atoms of a starting dicarboxylic acid.

(18) The polyether block amide material may comprise polyether segments and polyamide segments in a ratio of polyether to polyamide of from about 60:40 to about 40:60, preferably from about 60:40 to about 50:50.

(19) A gas sampling line having a channel for conducting respiratory gases traversing, in addition to the patient interface connector and the gas monitor connector, only the above-mentioned gas sampling tube provides, in addition to its functional performance in view of moisture and water removal as well as non-adsorption and non-absorption of gas components, further advantages: Manufacture, e.g. extrusion, of the gas sampling tube from only one starting material is considerably simpler than a process of manufacture involving joining tube portions of different materials together. Furthermore, a smooth inner surface of the channel, without material seams, may more easily be achieved by such a gas sampling line. Thus, the risk for signal distortion is decreased.

(20) It is, however, contemplated that parts of the channel for conducting respiratory gases may traverse tube sections of other materials, such as PVC or other conventional tube materials, included in the gas sampling line, in addition to the above-mentioned gas sampling tube. Such a design is preferred when a gas sampling line having a combination of different properties, as provided by different tube materials, is desired. It may also be a cost-effective solution to combine a lower priced tube material with the gas sampling tube, particularly for long gas sampling lines.

(21) It is advantageous when the gas sampling line further comprises a drying assembly comprising

a casing; and a hydrophilic member disposed within the casing and being in fluid contact with the channel,

wherein the casing is comprised of a second polyether block amide material, the polyether segments of which comprise polyethyleneoxide.

(22) Fluid contact between the hydrophilic member and the channel may be obtained, e.g., by surrounding a portion of the channel by a part of the hydrophilic member or by disposing a part of the hydrophilic member within a portion of the channel. When the gas sampling line comprises such a drying assembly, water or moisture present in respiratory gases conducted in the channel may be absorbed in, adsorbed to and/or stopped by the hydrophilic member. The hydrophilic member provides a wicking action, transporting moisture or water to the casing through which it is removed, thus not accumulating in the drying assembly. As described above in connection with the gas sampling tube, moisture and condensed water permeate through the polyether block amide material of the casing. Consequently, the drying assembly, and thus the entire gas sampling line, may be used for a prolonged period of time without change thereof.

(23) A drying assembly as disclosed above and further detailed below may also be applied in a conventional gas sampling line, i.e. a gas sampling line wherein the gas is conducted through a tube of a conventional tube material not permeable by moisture and/or water.

(24) The hydrophilic member may consist of a hydrophilic filter material having a large filtering area and being able to absorb moisture and water, e.g. in the form of sudden bursts of condensed water not yet removed during passage of the sampled gas through the gas sampling line. The drying assembly is typically positioned, along the gas sampling line, close to the end intended for connection to a gas monitor, e.g. adjacent to the gas monitor connector, in order to provide a final hinder to any water or moisture not removed during passage of the sampled gas through the gas sampling line.

(25) The polyamide segments of the second polyether block amide material may comprise polyamide-12, polyamide-11 or polyamide-12.12, preferably polyamide-12. The second polyether block amide material may comprise polyether segments and polyamide segments in a ratio of polyether to polyamide of from about 60:40 to about 40:60, preferably from about 60:40 to about 50:50. Further properties and advantages of second polyether block amide material resemble those of the first polyether block amide material and may be gathered above.

(26) The drying assembly may extend along the outside of a member defining the channel. This is an advantageous arrangement in that the hydrophilic member may transport moisture or water to a large area casing, thus allowing for efficient moisture and water permeation through the casing material, while maintaining a compact design of the gas sampling line. In particular, it is to be mentioned that the drying assembly may extend along the outside of the gas sampling tube as well as along the outside of tube sections of other materials, included in the gas sampling line.

Alternatively, it may be suitable to arrange the hydrophilic member within the gas sampling tube being comprised of the polyether block amide material. A portion of the gas sampling tube being comprised of the polyether block amide material may, in other words, surround the hydrophilic member, said portion of the gas sampling tube thus representing the casing of the drying assembly.

(27) The drying assembly may further comprise a hydrophobic member disposed across the channel. The hydrophobic member acts as a hydrophobic filter allowing the sampled gases to pass while protecting the gas monitor from undesirable substances or objects (e.g. bacteria or other bodily excretions) present in the sampled gas and hindering any remaining water from reaching the gas monitor. The hydrophobic member may or may not be positioned within the housing.

(28) Gas sampling lines according to the invention are typically single patient use disposables. The gas sampling lines may be adapted for a wide range of patient categories, such as infant, adult, or paediatric patients. It is particularly useful to provide a gas sampling line adapted for patients having a high breath rate, such as infants, and/or a weak respiration (i.e. providing a low gas flow in the gas sampling line), such as infants and/or pediatrics. The gas sampling lines may be adapted

to a wide range of applications, such as gas sampling from intubated patients, or nasal and/or oral gas sampling. Accordingly, in order to serve different patient categories or to allow use in different applications, the patient respiratory interface connector may be a nasal prong, a nasal cannula, an oral prong, a conical fitting, or a male or female Luer end. Thus, it is to be understood that the term “patient respiratory interface” may refer to an organ, e.g. the nose, of a patient as well as to a connection point in the respiratory loop of a mechanically ventilated patient. Although the gas sampling lines may be provided in any length, a typical length suited for practical use would be 1 to 3 m.

(29) Gas samples drawn through a gas sampling line according to the invention are typically intended for analysis in a gas monitor, such as a spectroscopic analyser. The gas monitor connector of the gas sampling line is preferably designed, together with the gas monitor receptacle, to provide a smooth gas path without a dead volume, to provide a safe no-break, no-leakage connection and to allow for a practically convenient handling of the patient, the sampling line and the gas monitor. Typical features provided by a preferred combination of the gas monitor connector of the sampling line and a gas monitor receptacle are an audible or otherwise perceptible “click” feedback assuring an operator that the gas monitor connector has correctly “clicked” into place in the gas monitor receptacle, a rotatable connection of the gas sampling line to the gas monitor, facilitating an operator's handling of the gas analysis system, absence of any intermediate connection elements, such as adapters, between the gas monitor connector and the gas monitor receptacle, thus providing a smooth gas path while avoiding risks for undesirable leakage, detection, e.g. by optical sensors, of a correct connection of the gas sampling line to the gas monitor and allowing, e.g., a sampling pump or the analyser to operate only when a correct connection is established, thereby improving the lifetime of components of the gas analysis system, and/or visual signals informing an operator of the operational status of the gas analysis system, e.g. green light when in operation, red light if occluded and blue light if an anesthetic agent is detected.

(30) In a second aspect of the invention, there is provided use of a tube comprised of a polyether block amide material, the polyether segments of which comprise polyethyleneoxide, for sampling of respiratory gases. Discussed above are advantageous compositions of the polyether block amide material.

(31) In a third aspect of the invention, there is provided a method for sampling of respiratory gases, the method comprising conducting respiratory gases through a tube comprised of a polyether block amide material, the polyether segments of which comprise polyethyleneoxide. Again, advantageous compositions of the polyether block amide material are discussed above.

(32) In a fourth aspect of the invention, there is provided a gas analysis system for analysing respiratory gases, comprising a gas sampling line as defined above and a gas monitor connectable to the gas sampling line. The gas analysis system may further comprise a respiratory device, such as a respirator, connectable to the patient respiratory interface.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) FIG. 1 is a schematic sectional view of a portion of a gas sampling line.

DETAILED DESCRIPTION OF AN EMBODIMENT OF THE INVENTION

(2) Referring to FIG. 1, some details of a an embodiment of a gas sampling line will be explained. The gas sampling line comprises a gas sampling tube 1, comprised of a material as specified above, connecting a patient respiratory interface connector (not shown) and a gas monitor connector 2. A gas monitor (not shown) may be coupled to the gas monitor connector 2. The gas sampling line further comprises a drying assembly comprising a housing 3, comprised of a material as specified above, and a hydrophilic member 4 disposed within the casing. The drying assembly 3, 4 extends

along the outside of the gas sampling tube 1. The hydrophilic member is in fluid contact with a channel 5 for conducting respiratory gases traversing the gas sampling line. A hydrophobic member 6 is disposed across the channel 5.

(3) When a sample of respiratory gases is conducted through the channel 5 of the gas sampling line towards the gas monitor connector 2, moisture or water present in the sample will be adsorbed to or absorbed by the hydrophilic member 4. A wicking action of the hydrophilic member 4 will transport the moisture or water towards the housing 3. Subsequently, moisture or water will permeate the housing 3 and be removed into surrounding air. The respiratory gases will pass the hydrophobic member 6 on their way towards the gas monitor connector 2, whereas undesirable objects or substances (e.g. bacteria, body excretions) will be withheld by the hydrophobic member 6 and not reach the gas monitor. The hydrophobic member 6 also serves as an additional measure to stop water or moisture from reaching the gas monitor.

(4) It is to be understood that the invention is not to be limited to the disclosed embodiment, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.

Examples

(5) Characteristics and advantages of the present invention are further illustrated by the following, non-limiting, examples.

(6) Preparation of Gas Sampling Tubes According to the Invention

(7) Gas sampling tubes according to the invention were prepared by extrusion of a polyether block amide material (available from Atofina under the trade name Pebax®) to form a tubing having an inner diameter of 1 mm and an outer diameter of 2.5 mm, and subsequent cutting of the tubing to obtain gas sampling tubes having a length of 2 m. The composition of the polyether block amide material is shown in Table 1.

(8) TABLE-US-00001 TABLE 1 Gas sampling tube according to the present invention Tube no. Composition 1 polyether block amide material of 55% polyethyleneoxide and 45% polyamide-12 Preparation of Comparative Gas Sampling Tubes

(9) As comparative gas sampling tubes were used 2 m lengths of four different tubings having an inner diameter of 1 mm and an outer diameter of 2.5 mm. The composition of the tubing materials and the construction of the tubings are shown in Table 2. Comparative tubes 3-5 represent well-known embodiments of gas sampling tubes used in respiratory care for the conduction of gases to gas analysis equipment.

(10) TABLE-US-00002 TABLE 2 Comparative gas sampling tubes Tube no. Composition and construction 2 polyether block amide material of 53% polytetramethylene oxide and 47% polyamide-12 (extruded) 3 polyvinylchloride (extruded) 4 polyvinylchloride/polyethylene (co-extruded; inner layer PVC, outer layer PE) 5 polyvinylchloride + Nafion® (1.9 m PVC joined to 0.1 m Nafion®)

Test Methods

(11) All tests were performed at room temperature of about 22° C. at a gas flow of 50 ml/min through the gas sampling tube. In the tests, tube no. 5 was arranged so that in the direction of the gas flow, the Nafion® portion was upstream of the PVC portion. A) Moisture test: A gas sample of moist air was passed from a simulated patient circuit, equipped with a heated humidifier, through a gas sampling tube to a water trap having a volume typical for disposable water traps for analysis of respiratory gases. The moist air leaving the simulated patient circuit had a relative moisture of 95-100% at 35-37° C. Water condensed in the tube was collected in the water trap and the time until the water trap had been filled with 200 µl liquid was recorded. B) Water test: A gas sample comprising dry air and drops of water was passed from a simulated patient circuit, equipped with a syringe pump for delivery of water, through a gas sampling tube to a water trap having a volume typical for disposable water traps for analysis of respiratory gases. The syringe pump was set to deliver one droplet of water per minute, corresponding to 100 µl liquid per hour. The liquid was

collected in the water trap and the time until the water trap had been filled with 200 μ l liquid was recorded. C) CO.sub.2 accuracy: The sampling tube was connected between an equipment providing alternating two reference gases (5% CO.sub.2 balanced N.sub.2 and synthetic air) according to EN ISO 21647:2004 (Medical electrical equipment—Particular requirements for the basic safety and essential performance of respiratory gas monitors), figure 102, and a gas monitor. The measuring equipment was set to alternate the reference gases at a frequency corresponding to 40 breaths per minute. The ratio of CO.sub.2 concentration measured by the gas monitor and CO.sub.2 concentration of the reference gas was recorded. D) Halothane accuracy: The sampling tube was connected between an equipment providing alternating two reference gases (5% CO.sub.2, 5% halothane balanced N.sub.2 and synthetic air) according to EN ISO 21647:2004, figure 102, and a gas monitor. The measuring equipment was set to alternate the reference gases at a frequency corresponding to 40 breaths per minute. The ratio of halothane concentration measured by the gas monitor and halothane concentration of the reference gas was recorded.

Results

(12) The results are shown in Table 3. The gas sampling tube according to the invention (tube no. 1) provided an outstanding combination of desirable results in the moisture and water tests as well as in the CO.sub.2 accuracy and halothane accuracy tests.

(13) TABLE-US-00003 TABLE 3 Results A) B) C) D) Moisture test Water test CO.sub.2

Halothane Tube no. (h)	(h) accuracy	accuracy 1	>24	>24	0.98	0.98	2	2.5	2	0.98	0.98	3	2.5	2	0.98
0.73	4	2.5	2	0.98	0.98	5	>24	2	0.97	0.74					

Claims

1. A gas sampling line having a channel for conducting respiratory gases comprising: a gas sampling tube, wherein at least a portion of the gas sampling tube comprises a polyether block amide material; a hydrophilic member disposed within the gas sampling tube surrounding at least a portion of the channel; a patient respiratory interface connector adapted to couple the gas sampling line to a patient respiratory interface; and a gas monitor connector adapted to couple the gas sampling line to a gas monitor.
2. The gas sampling line of claim 1, wherein a second portion of the gas sampling tube comprises a material that is not permeable to moisture and/or water.
3. The gas sampling line of claim 1, wherein the hydrophilic member is in contact with the portion of the gas sampling tube comprising a polyether block amide material.
4. The gas sampling line of claim 1, wherein the polyether block amide material comprises polyether segments and polyamide segments, and the polyether segments comprise polyethyleneoxide.
5. The gas sampling line of claim 4, wherein the polyamide segments of the polyether block amide material is selected from the group consisting of polyamide-12, polyamide-11, and polyamide-12.12.
6. The gas sampling line of claim 4, wherein the polyether segments and the polyamide segments in the polyester block amide material has a ratio from about 60:40 to about 40:60.
7. The gas sampling line of claim 1, further comprises a hydrophobic member disposed across the channel.
8. A method for sampling of respiratory gases, the method comprising conducting respiratory gases through the gas sampling line of claim 1.
9. The method according to claim 8, wherein the polyamide segments of the polyether block amide material is selected from the group consisting of polyamide-12, polyamide-11, and polyamide-12.12.
10. The method according to claim 8, Wherein the first polyester block amide material comprises polyether segments and polyamide segments in a ratio of polyether to polyamide from about 60:40

to about 40:60.

11. A gas analysis system for analyzing respiratory gases, comprising the gas sampling line as defined in claim 1.

12. The gas analysis system according to claim 11, further comprising a respiratory device connectable to the patient respiratory interface.
